

Detection of Acute Lymphoblastic Leukemia Subtypes using YOLO and Mask R-CNN

Chastine Fatichah
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
chastine@if.its.ac.id

Tanzilal Mustaqim
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
derekln14@gmail.com

Nanik Suciati
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
nanik@if.its.ac.id

Aldinata Rizky Revanda
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
aldinatarizky@gmail.com

Abstract— White Blood Cell (WBC) detection and counting on Microscopic Blood Cell images automatically can support expertise in diagnosing Acute Lymphoblastic Leukemia (ALL) more easily and quickly. Previous research commonly uses conventional approaches for ALL subtype detection that needs several stages, such as WBC segmentation, touch cell separation, feature extraction, and classification. We present object detection and instance segmentation techniques that require only one learning framework without needing separate stages. In this paper, we compare the performance of the YOLO and Mask R-CNN models for the detection of ALL subtypes using evaluation metrics such as precision, recall, F1, and mAP. The results show that the YOLOv4 outperforms YOLOv5 and Mask R-CNN in the detection of ALL subtypes. The YOLOv4 model has slightly better performance than the YOLOv5 model and Mask R-CNN, with an F1 value of 89.5% and a mAP value of 93.2%, respectively.

Keywords—Acute Lymphoblastic Leukemia image, Object detection, instance segmentation, YOLO, Mask R-CNN

I. INTRODUCTION

Leukemia is typically identified by studying microscopic blood images. For the examination, the hematologist can conduct a manual analysis. Each white blood cell's morphological features are examined by the hematologist in order to determine the type of leukemia that is present. If Acute Lymphoblastic Leukemia (ALL) supports the diagnosis, additional tests are performed on the morphology of ALL of the L1, L2, and L3 subtypes. However, it takes a lot of time and effort to analyze white blood cells manually [1]. This manual procedure is heavily reliant on the pathologist's abilities and experience, which has the potential to lead to issues owing to erroneous results [2]. Hematology counters are automated devices that do white blood cell analysis fast and automatically today, but they are still so expensive that not all hospitals, let alone entire nations, can afford them [3]. Consequently, a computerized system is required to assist in the analysis of white blood cells, particularly lymphoblasts.

Segmentation, separating touched cells, feature extraction, and classification are typically required steps in computer-assisted diagnostic exams [4]. Many researchers have used traditional machine learning techniques. Conventional approaches, such as conventional digital image processing, can be helpful, but the process will be more

difficult because segmentation, feature extraction, and classification are all connected steps [5]. As a result, depending on the state of the input image, the conventional technique necessitates more steps [6]. Additionally, employing conventional technique approaches will be time-consuming for accurate nuclei detection, boundary separation to identify overlapping cells, RoI segmentation, exact feature extraction, and optimal feature selection [7]. The success of each stage, particularly the stage that separates cells that are in contact, is crucial to the ALL subtype discovery procedure.

Recently, the medical industry has made extensive use of the object detection method. The object detection approach performs exceptionally well and is faster and more accurate than traditional methods. The object detection method only needs one short step to detect items and doesn't involve many processes. It is possible to study the morphology and placement of ALL subtypes using an object detection or instance segmentation strategy.

The YOLO object detection model uses deep learning to accomplish object detection tasks inside a single learning framework [8], [9]. Yolov4 was utilized in research by [10] to find white blood cells and assess whether they were ALL or normal. The research [11] propose modify of YOLOv4 for detection of ALL subtypes in Blood Microscopic images. Previous research has used YOLO to identify diseases, including cancer. YOLOv3 is used to classify and localize significant areas in biopsy images in order to detect prostate cancer [12]. In two independent datasets, the study's findings can provide detection accuracy scores of 97% and 89%. A different study [13] examined Yolov1, Yolov2, and Yolov3 to identify melanoma using the best mAP measures. According to the comparative results, Yolov2 has the highest mAP value (83%), followed by Yolov1 (37%), and Yolov3 (77%). In order to find breast cancers in mammography and ultrasound datasets, the Yolo model was developed into Yolov4 and Yolov5. The Yolov3 and Faster R-CNN models were outperformed by both models [14, 15].

A simple, adaptable, and all-encompassing framework for object instance segmentation is the Mask R-CNN [16]. Mask R-CNN is an effective tool for accurately detecting and classifying items in an image and can simultaneously provide a high-quality segmentation mask for each object [16, 17]. Mask R-CNN can distinguish between different objects in an image, even those that overlap objects. [18] suggests the prior

study on ALL subtype segmentation on White Blood Cell images using Mask R-CNN. Before applying the Mask R-CNN for instance segmentation, they also propose an enhancement technique to improve the accuracy.

The objective of our research is to compare the performance between YOLO and Mask R-CNN methods for detecting ALL subtypes. For YOLO, we also compare the performance of YOLOv4 and YOLOv5. Comparisons were made using the values of evaluation metrics such as precision, recall, F1 and mAP, which were derived from detection measurements performed on the L1, L2 and L3 subtypes. The datasets used have the same source and composition for ALL subtypes, so that the comparison process is not skewed towards one model.

The remaining papers are arranged as follows: Section II presents the literature review such as YOLO and Mask R-CNN. Section III discusses the research methodology, datasets, and evaluation techniques for the suggested methodologies. Section IV discusses the results and discussion. The research findings' conclusions and potential directions for further work are discussed in Section V.

II. LITERATURE REVIEW

A. YOLO

YOLO, which stands for You Only Look Once, is an object detection algorithm that can work in real time [19]. YOLO performs a multi-step process automatically by combining the classification and bounding box prediction processes for detected objects using a single neural network. Because of this, YOLO can classify and detect objects significantly faster than two separate neural networks. Conventional image classifiers are reused to identify bounding box objects by regression. This method predicts bounding boxes and class labels for each individual boundary by training a single neural network from start to finish. This model divides the image into several grid cells, each of which predicts a bounding box. The x,y coordinate system for each grid cell predicts a bounding box based on the width, height, and confidence values.

One of the main problems that must be addressed in YOLOv1 is the detection of smaller objects in the image. This issue was addressed in YOLOv2, which splits an image into a smaller 13x13 grid cell than the previous version. YOLOv3 introduces a number of improvements, including the use of multi-label classification in place of the SoftMax function [8]. Using multi-label classification, YOLOv3 can classify objects semantically, such as people and men, so that one object can be classified with several classes based on the correct classification.

To increase accuracy, YOLOv4 makes a number of network enhancements [20]. The YOLOv4 design has three YOLO heads that are CSPDarknet53 backbone network, SPP (Spatial Pyramid Pooling), and PANet for the neck. The variables on YOLOv4 represent the bounding box attributes, which include center, width, and height coordinates as well as related objectivity scores.. Different functions are carried out by the backbone, neck, and head architectural elements in YOLOv4. In order to extract map features from the image, the CSPdarknet53 backbone is used. An SPP that performs maximum pooling with four separate pools is located on the neck; PANet will integrate and process the output from these four processes. By using low-level and high-level feature

maps for upsampling and downsampling, PANet aims to increase the quality of feature maps. Using three distinct sizes, object detection is done in the head region. The first YOLOv4's head section is used to detect little items, the second YOLOv4's head section is used to detect medium-sized objects, and the third YOLOv4's head section is used to detect large objects. Glenn Jocher introduced YOLOv5 utilizing the Pytorch framework shortly after YOLOv4 was made available. YOLOv5 can enhance the object detection model's capability to detect small objects in autonomous vehicles [21].

The YOLO model is utilized in this study due to its high level of precision and real-time object detection capabilities. In contrast to other object detection models, the YOLO model can detect objects without the need for multiple separate steps. In a single learning framework, the Yolo model can detect objects in the image. YOLOv4 and YOLOv5 are the models utilized in this study.

B. Mask R-CNN

Mask Region-based Convolutional Neural Network (Mask R-CNN) is an expansion method for Faster Region-based Convolutional Neural Network (Faster R-CNN) that predicts masking on objects in parallel with existing branches for boundary box detection [16]. At the pixel level, the objective is to separate classification and prediction masking tasks. The Mask R-CNN was created to correct misaligned object segmentation in the Faster R-CNN algorithm. This is accomplished by substituting the RoIPooling method used in the Fast R-CNN and Faster R-CNN algorithms with the RoIAlign method.

Mask R-CNN operation begins with the Region Proposal Network module, which searches for potential objects in an image using RoIAlign. The possible positions of objects in the image are then assigned a boundary and a region of interest is identified (ROI). RPN then inputs sub-images of potential objects of varying sizes into a Convolutional Neural Network, which produces a set of rectangles with object similarity scores. Class prediction (classification) and box offset (regression) were performed in parallel on possible objects based on the outcomes of the previous module. The Mask R-CNN also outputs a binary mask for each RoI at this stage.

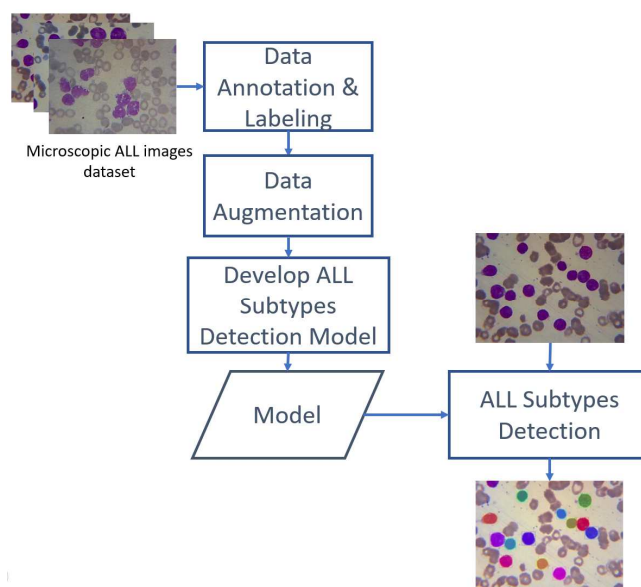


Fig. 1. Block diagram of ALL Subtypes Detection

In addition to the existing branches for classification and localization, Mask R-CNN receives a third branch to predict the masking of objects. Masking branches are small, fully connected networks that are applied to each Region of Interest (RoI), thereby predicting pixel-by-pixel segmentation. In contrast to semantic segmentation, additional masking permits the intelligent selection of pixels on each object segment and the extraction of each object separately without background (instance segmentation). The result of the Mask R-CNN method is the boundary box and segmentation mask for each object leading to a segmentation instance.

The Mask R-CNN is used as a comparison model to YOLO due to its ability to detect objects through instance segmentation. Nevertheless, the Mask R-CNN model's detection speed is slower than YOLO's. In this study, the Mask R-CNN model was used to compare its performance in detecting ALL subtypes to that of YOLO.

III. RESEARCH METHOD

Fig. 1 displays the block diagram of ALL subtypes detection. In this section, we presents dataset description, ALL cells annotation and labeling, Data augmentation, ALL subtypes detection using YOLO, and ALL subtypes detection using Mask R-CNN.

A. Dataset Description

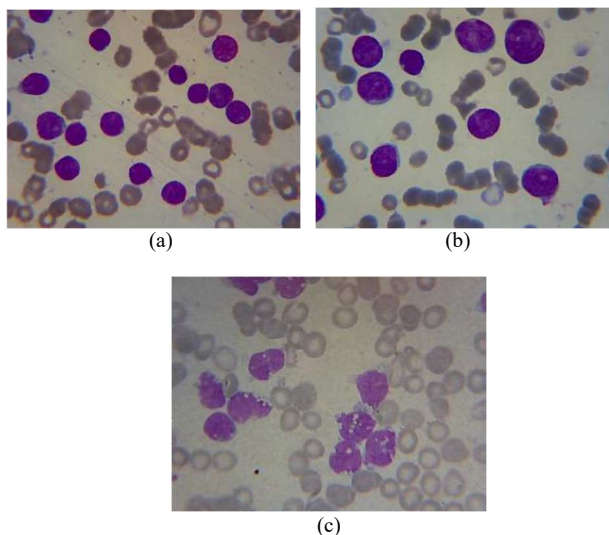


Fig. 2. ALL images (a) L1 subtype (b) L2 subtype (c) L3 subtype

White blood cell microscopy images from patients with ALL make up the whole dataset used in this study, hence there are no images of healthy patients [22]. There are 301 multi-cell ALL images totaling data, of which 128 are L1 type images, 63 are L2 type images, and 110 are L3 type images. The three forms of ALL in the dataset are represented by white blood cell microscope images in Fig. 2.

B. Data Augmentation

Image data augmentation is a method for generating new images from existing image data. The data augmentation process makes minor adjustments to the image, such as adjusting the image's brightness, rotating it, or shifting the image's subject horizontally or vertically.

Image augmentation techniques enable the model to artificially increase the size of the training set, thus providing

the model with additional training data. This increases the model's ability to recognize new variant representations from augmented image data, thereby increasing the model's evaluation metric value. In order to improve the performance of the model, augmentation techniques are also beneficial for reducing the level of overfitting.

Rotate, zoom, and flip are the standard data augmentation techniques utilized in this study to generate new image variations. By rotating the original image a few degrees, the rotation method generates a different image in which the appearance of the image's objects can vary. The zoom method enlarges the original image in order to obtain a range of object sizes. The flip method creates a different image by inverting the image's orientation, allowing the model to recognize image objects from a different perspective.

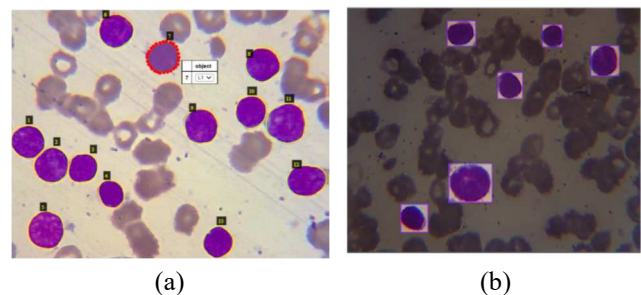


Fig. 3. ALL annotation and labeling (a) Mask R-CNN (b) YOLO [18]

C. ALL Cells Annotation and Labeling

Prior to using YOLO and Mask R-CNN, microscopic images annotation and labeling for ALL subtypes is completed. Fig. 3 (a) and Fig. 3 (b) show the annotation and labeling for the Mask R-CNN and YOLO, respectively. Due to the differences in output between YOLO (label and bounding box) and Mask R-CNN (label and object contour), we need two separate way of annotation and labeling techniques.

D. ALL Subtypes Detection using Mask R-CNN

The Mask R-CNN model is utilized for segmentation of ALL subtypes yielding predicted classes, bounding boxes, and contour objects. The training data are used to construct the Mask R-CNN model, while the testing data are utilized to evaluate the model's performance. The contour of cells can identify the area of cells that have been touched. Fig. 4 demonstrates the segmentation of ALL subtypes using Mask R-CNN.

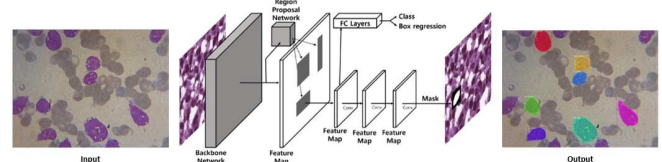


Fig. 4. ALL subtypes segmentation using Mask R-CNN

E. ALL Subtypes Detection using YOLO

There are two YOLO methods that are applied for ALL subtypes detection i.e. YOLOv4 and YOLOv5. We compare the performance between both methods using Precision, Recall, F1 score, and mAP. The YOLO models are build from the training data, and the testing data is used to evaluate the performance of YOLO models. The output of YOLO models are predicted classes and bounding boxes. The ALL subtypes segmentation using YOLO is illustrated in Fig. 5.

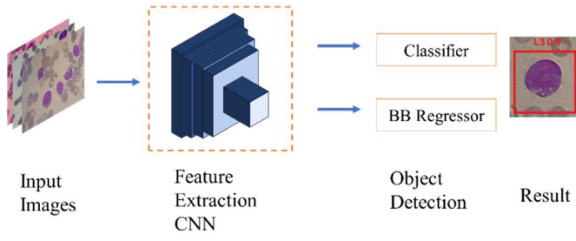


Fig. 5. ALL subtypes segmentation using YOLO

F. Evaluation Methods

Mean Average Precision (mAP), Recall, F1, and Precision are used as evaluation metrics in this study. Comparing the overall positive anticipated model results to the genuine positive prediction is the purpose of the precision measurement technique, see (1). Equation (2), which deals with recall, gives information for comparing genuine positive predictions to the specific data that made the true positive prediction. F1 calculates the percentage of the average recall and precision values (3).

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (1)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (2)$$

$$\text{F1} = \frac{2 \times (\text{Recall} \times \text{Precision})}{\text{Recall} + \text{Precision}} \quad (3)$$

Average precision (AP) is a common metric for assessing the performance of object detection models. AP allows avoiding any bias from a single measurement point relative to the entire measurement. In addition, the evaluation metrics of precision and recall are insufficient for measuring the overall performance of the model in detecting image objects. AP is the primary factor used to compare the performance of the YOLO and Mask R-CNN models. AP is the average precision across all distinct recall levels. The formula for average precision is shown in equation (4). Equation (5) illustrates how to obtain the mAP by first calculating the AP for each class and then averaging the outcomes across several classes.

$$\text{AP} = \sum_{k=0}^{k=n-1} [\text{Recalls}(k) - \text{Recalls}(k+1)] * \text{Precisions}(k) \quad (4)$$

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (5)$$

IV. RESULTS AND DISCUSSION

In this Section, the experimental results on ALL subtypes are presented. There are three subsection, i.e. experiments on ALL subtypes using YOLO method, experiments on ALL subtypes using Mask R-CNN, and comparison results between YOLO and Mas R-CNN. We also discuss the analysis of the results especially the missed detection of ALL objects.

A. Experiments on ALL Subtypes using YOLO

Fig. 6 displays the visual results of the YOLOv4 and YOLOv5 model experiment. Fig.6 (b) demonstrates that, according to the results of YOLOv4, there are still two false positives from the L2 subtype that should be L1. Fig. 6 (c) depicts the experiment results from the YOLOv5 model, which produced four false positive subtypes of L2. In this instance, the YOLOv4 model is superior to the YOLOv5

model in analyzing each cell's representation of ALL subtypes, especially L1 and L2. The morphology of the L1 and L2 subtypes are comparable, particularly on the sides of cell objects that touch and overlap, so the model's performance is not optimal.

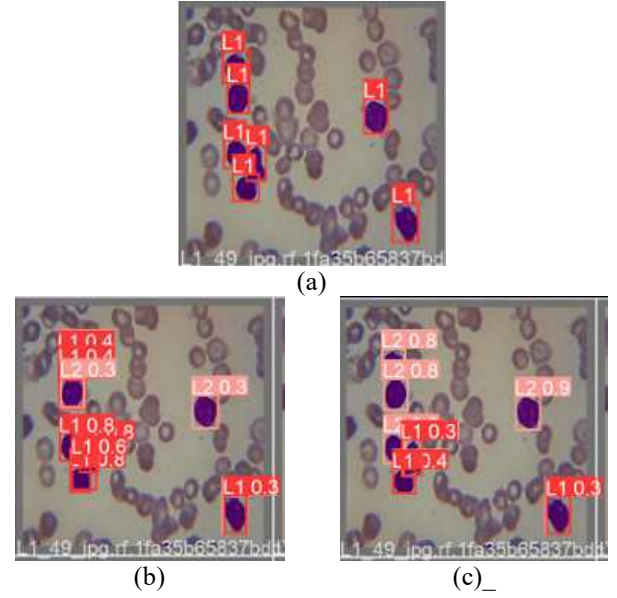


Fig. 6. The result of ALL subtypes detection using YOLO (a) YOLOv4 (b) YOLOv5

B. Experiments on ALL Subtypes using Mask R-CNN

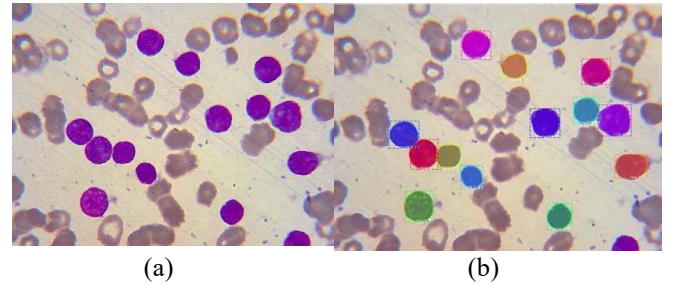


Fig. 7. The result of ALL subtypes detection using Mask R-CNN

The visual experiment sample results from the Mask R-CNN model are displayed in Fig. 7. The Mask R-CNN model was capable of performing instance segmentation on the majority of ALL subtype cell objects within the image. Despite this, there are still a number of objects that have not been detected among the cases of included objects. The Mask R-CNN Model has advantages when detecting L3 but limitations when detecting object subtypes L1 and L2. As with the YOLO model, objects L1 and L2 have morphological characteristics that are nearly identical, rendering the model ineffective.

C. Comparison Results using YOLO and Mask R-CNN

TABLE I. EXPERIMENTAL RESULTS ON ALL SUBTYPES DETECTION

Cell Types	Evaluation Method	Yolov4	Yolov5	Mask R-CNN
L1	Precision	89.6%	93.0%	81.8%
	Recall	95.0%	93.3%	90.4%
	F1	92.1%	92.9%	85.9%
	mAP	96.3%	96.8%	82.5%

L2	Precision	87.1%	79.0%	72.5%
	Recall	77.6%	79.5%	52.7%
	F1	81.9%	75.4%	58.9%
	mAP	85.4%	83.6%	70.0%
L3	Precision	91.7%	93.2%	92.7%
	Recall	96.0%	92.9%	93.5%
	F1	93.8%	92.9%	93.6%
	mAP	97.7%	97.9%	93.3%
Global	Precision	89.5%	88.4%	79.9%
	Recall	89.5%	88.6%	79.4%
	F1	89.5%	88.3%	79.5%
	mAP	93.2%	92.7%	81.9%

Table I displays the results of a comparison of evaluation metrics from the YOLOv4, YOLOv5, and Mask R-CNN model testing processes. The results of the evaluation metrics are divided into four categories, each with four metrics. To avoid bias caused by the number of false positives and true positives detected in precision and recall values, the comparison procedure emphasizes the F1 and mAP values. Evaluation of the L1 subtype category reveals that the YOLOv5 model outperforms the F1 and mAP values, but differs from YOLOv4 by less than 1%. The YOLOv4 model has the highest recall rate, but its recall value is lower than that of the YOLOv5 model, indicating that many false positives are still identified. The Mask R-CNN model still has F1 and mAP values below 90%, indicating that a substantial number of undetected true positives and false positives have been detected.

In the L2 category, the YOLOv4 model outperforms the YOLOv5 model and the Mask R-CNN model, in contrast to the L1 category, where the YOLOv5 model and the Mask R-CNN model perform better. However, the YOLOv4 model's recall value is 1.9% less than that of the YOLOv5 model due to a number of undetected true positives. The overall results of the Mask R-CNN model for the L2 category are inferior to those for L1. This is because objects of subtypes L1 and L2 share similar characteristics. Due to the small number of L2 datasets compared to other class objects, the mAP values of all models are lower than those of the L1, L3, and Global models, so the model performance is not optimal.

In the L3 category, the YOLOv4 model produced the highest F1 value, whereas the YOLOv5 model produced the highest mAP value. The Mask R-CNN model remains inferior to the two YOLO models. Nevertheless, compared to the previous class, the Mask R-CNN model has the highest evaluation metric value, with a value greater than 90%. This is because the Mask R-CNN model has distinct characteristics from the L1 and L2 object subtypes, allowing it to function optimally.

In the global category, the category-wide average is used to determine the model with the highest performance. The YOLOv4 model has higher F1 and mAP values than the YOLOv5 model and the Mask R-CNN model, according to the average results. However, the YOLOv4 model still has a recall value and precision below 90%, with some ALL subtype cell objects detected as false positives and others not detected.

V. CONCLUSIONS

In clinical applications, a precise classification of ALL must be developed in order to assist pathologists in performing a rapid and accurate diagnostic procedure. Particularly in the lymphoblast detection, segmentation, and classification

processes, it is still challenging to determine the optimal approach for this task. For the detection of ALL subtypes, we propose the use of object detection and instance segmentation techniques. YOLOv4 outperforms YOLOv5 and Mask R-CNN in detecting ALL subtypes. YOLOv5 has the best performance for detecting L1 subtypes. The Mask R-CNN detects ALL subtypes with the lowest performance. To improve model performance in future work, data augmentation can be used to add data and modify the YOLO and Mask R-CNN models, particularly to overcome overlapping cells, small cells, and touching cells.

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